MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU -UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

(2024 Admission onwards)

Minor Course – MG4DSCBCH202 – Metabolism of Carbohydrates, Proteins and Lipids

Duration: 1 Hour 30 Minutes Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple Choice Questions
(Answer all questions; each question carries 1marks)

1. The key regulatory enzyme of glycogenolysis is [U][CO2]
A) Glycogen synthase B) Hexokinase C) Glycogen phosphorylase D) Glucose-6-phosphatase

2. Pancreatic amylase acts in the [K][CO1]
A) Stomach B) Mouth C) Small intestine D) Large intestine

3. The coenzymes required for the action of glutamate dehydrogenase [K][CO3]
A) NAD B) NADP C) both A and B D) FAD and FMN

4. Decarboxylation of amino acids results in the formation of [U][CO3]
A) Keto acids B) Amines C) Amides D) Alcohols

5. Which of the following is a component of the fatty acid desaturase system? [U][CO4]
A) Cytochrome c B) Cytochrome b_5 C) Cytochrome a_3 D) Cytochrome

(1x 5 = 5 marks)

Part B
Fill in the Blanks
(Answer all questions; each question carries 2 marks)

6. The complete aerobic oxidation of one molecule of glucose yields approximately…. [U][CO2] ATP, whereas anaerobic oxidation yields only …….ATP

7. …… helps to transport fatty acids to mitochondria for degradation [K][CO4]

8. The final electron acceptor in the ETC is……., which combines with hydrogen ions to [U][CO2] form…….

9. Alkaptonuria is caused by deficiency of the enzyme….,leading to accumulation of…… [U][CO3]

10. Digestion of carbohydrates begins in the mouth with the enzyme…… [U][CO1]

(2 x 5 = 10 marks)

Part C
Short Answer Questions
(Answer any 5 questions; each question carries 3 marks)

11. Describe the different fates of pyruvate in the body? [E[[CO2]
12. What is gluconeogensis? [U][CO2]
13. Explain the deamination reaction. [K][CO3]
14. Discuss the significance of carnitine shuttle. [U][CO4]
15. Name the three ketone bodies? [U][CO4]
16. Illustrate the significance of glucogenic and ketogenic amino acids. [E][CO3]
17. Describe how lipids are digested and absorbed in human body. [U][CO1]

(3 x 5 = 15 marks)

Part D
Short Essay Type Questions
(Answer any 4 questions; each question carries 5 marks)

18. Discuss the metabolic pathway of glycolysis. [E][CO2]
19. What is citric acid cycle? Analyze the steps involved? [An][CO2]
20. Explain the steps involved in urea cycle. [U][CO3]
21. Elaborate on β oxidation and its energetics. [E][CO4]
22. Analyze the process of fatty acid synthesis [An][CO4]
23. Describe the digestion and absorption of proteins. [K][CO1]

(5x 4 = 20 marks)